

JUG BAY, MD, USA

WMO: 997994

Lat: 38.781N

Lon: 76.708W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -7.4	(d) -19.0	(e) 0.7	(f) -8.2	(g) -15.4	(h) 1.0	(i) -4.1	(j) 7.5	(k) 4.8	(l) 7.0	(m) 2.6	(n) 2.0	(o) 310	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	34.1	25.5	32.7	24.8	31.4	24.1	26.7	32.1	25.8	31.0	25.2	30.1	1.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.3	29.2	24.5	19.4	28.4	23.8	18.6	27.8	83.0	32.2	79.8	31.1	76.7	30.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.6	(b) 5.6	(c) 4.9	(e) -12.3	(f) 37.0	(g) 2.9	(h) 2.2	(i) -14.4	(j) 38.6	(k) -16.1	(l) 39.9	(m) -17.7	(n) 41.1	(o) -19.7	(p) 42.8		
(4) 6.6	(5) 5.6	(6) 4.9	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	1.0	3.4	6.9	13.7	19.0	23.3	25.7	24.3	21.2	14.5	8.1	5.2
	DBStd	9.39	5.14	4.72	5.11	4.70	4.26	3.08	2.53	2.32	3.49	4.22	4.39	4.91
	HDD10.0	878	282	193	124	14	1	0	0	0	0	10	87	166
	HDD18.3	2372	537	419	354	154	47	2	0	0	12	131	308	407
	CDD10.0	2306	3	8	29	126	279	398	486	443	336	150	30	17
	CDD18.3	758	0	0	1	16	67	150	228	184	98	13	1	0
	CDH23.3	6594	0	0	12	186	552	1340	2169	1497	738	97	3	0
	CDH26.7	2529	0	0	1	59	151	515	959	573	251	21	0	0
Wind	WSAvg	1.7	2.1	2.3	2.3	2.0	1.7	1.3	1.1	0.9	1.1	1.4	1.8	1.9
Precipitation	PrecAvg	1132	79	69	98	90	100	113	109	109	103	89	71	92
	PrecMax	1609	166	178	238	143	263	260	189	266	275	207	165	184
	PrecMin	847	33	13	0	47	28	51	46	35	1	4	11	37
	PrecStd	158	30	37	46	28	52	55	40	49	71	52	43	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.6	21.3	24.8	32.0	32.2	35.3	36.9	34.6	33.6	29.2	23.3	20.9
		MCWB	13.0	14.5	16.6	18.6	24.0	26.2	26.1	24.9	24.5	21.4	16.8	17.9
	2%	DB	14.9	17.1	21.5	28.2	30.1	33.0	34.7	32.8	31.8	25.9	19.7	18.5
		MCWB	11.0	11.5	15.3	19.0	22.9	25.1	25.9	24.6	23.9	19.1	14.6	15.7
	5%	DB	11.4	13.9	18.4	25.2	28.4	31.4	33.0	31.4	29.7	23.6	17.3	15.5
		MCWB	7.9	8.9	13.3	17.7	21.4	24.2	25.3	23.8	23.1	18.3	13.8	13.2
	10%	DB	8.8	10.8	15.3	22.5	26.8	29.8	31.5	30.1	27.7	21.6	15.2	12.9
		MCWB	6.0	6.6	10.6	15.9	20.5	23.3	24.6	23.1	21.7	17.4	12.1	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.8	18.0	21.6	25.0	27.2	28.4	26.9	25.8	22.8	18.4	18.9
		MCDB	18.2	20.3	23.1	25.5	31.4	33.3	33.9	32.1	31.2	27.1	20.9	20.5
	2%	WB	11.5	13.0	15.9	19.9	23.9	26.1	27.0	25.6	25.0	21.4	16.3	15.9
		MCDB	14.2	17.0	20.1	24.7	29.2	31.5	32.3	30.4	29.8	23.8	18.7	17.7
	5%	WB	8.8	10.0	14.1	18.0	22.8	25.2	26.3	24.9	24.2	20.3	14.6	13.9
		MCDB	10.7	12.9	18.3	23.1	27.4	30.5	31.7	29.3	28.5	22.9	16.8	15.3
	10%	WB	6.2	7.8	11.5	16.4	21.7	24.3	25.6	24.1	23.3	18.3	12.7	10.8
		MCDB	8.5	11.1	14.2	20.7	25.5	28.7	30.6	28.0	26.4	21.1	14.8	12.5
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	10.5	11.9	11.6	11.5	11.4	11.3	11.3	10.9	10.5	9.1
		MCDBR	13.2	15.0	15.3	15.2	12.9	13.2	13.0	12.6	13.3	13.7	13.3	11.6
		MCWBR	10.0	9.6	9.5	6.9	6.2	6.0	5.3	5.1	5.5	7.0	8.8	9.2
	5% WB	MCDBR	12.8	13.7	14.1	12.2	11.7	12.0	12.2	10.8	10.9	10.8	11.6	10.8
		MCWBR	10.9	9.9	9.7	7.2	5.8	5.6	5.3	5.0	5.1	6.9	8.8	9.2
Clear-Sky Solar Irradiance	taub	0.309	0.321	0.356	0.397	0.438	0.492	0.517	0.509	0.420	0.361	0.325	0.310	
	taud	2.495	2.449	2.407	2.310	2.226	2.121	2.067	2.075	2.297	2.466	2.525	2.531	
	Ebn at Noon	870	906	902	880	847	799	774	769	823	848	845	843	
	Edh at Noon	78	93	108	126	140	156	163	158	119	91	76	71	
All-Sky Solar Radiation	RadAvg	2.12	2.96	3.94	4.98	5.63	6.23	6.10	5.43	4.42	3.31	2.41	1.86	
	RadStd	0.17	0.32	0.31	0.37	0.65	0.47	0.30	0.30	0.49	0.36	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+77

Nomenclature: See separate page